Stage-Gate Incentive Design in Product Development With Resource Competition Among Parallel Projects

Xiaoxiao Li, Binghan Kou, and Jie Gao

Abstract—**Firms often follow the stage-gate process to manage multiple projects in product development and delegate stages of idea discovery and product development to project managers. This framework challenges the firm in designing efficient investment schemes and compensation contracts for the managers due to information asymmetry and resource conflicts among the projects. To address these challenges, we develop a principal-agent model by considering two projects in the presence of asymmetric idea quality information and unobservable development effort. Our results reveal that to incentivize the managers to report truthful idea quality and exert proper development effort, the firm should terminate low-quality ideas even though these ideas can bring positive returns with proper investment. Moreover, as the idea quality decreases, the firm should provide a fixed-salary contract rather than an outcome-based one. The asymmetric idea quality information can create underinvestment inefficiency, and unobservable efforts can exacerbate this inefficiency. We further provide conditions under which encouraging resource competition among parallel projects helps alleviate the underinvestment issue. Overall, our results provide guidelines for designing investment schemes and compensation contracts for parallel research and development (R&D) projects in the presence of moral hazard and adverse selection and shed light on managing the resource competition to improve investment efficiency.**

Managerial Relevance Statement—**This study addresses the challenges firms face in designing effective investment schemes and compensation contracts within the context of stage-gate process with information asymmetry and resource conflicts across parallel projects. Specifically, the firm should terminate low-quality ideas, while continuing medium- and high-quality ideas with low and high investment, respectively. For compensation contracts, as the idea quality increases, we recommend transitioning from fixed-salary contracts to outcome-based linear contracts by providing more unit bonuses but reduced fixed salaries. Moreover, we highlight the critical need for firms to allocate less investment than the first-best level at the gate in early periods, and guarantee a higher investment than the first-best level at the gate in later periods. Finally, to mitigate this inefficiency and improve the profit, we recommend the following plans by adjusting the resource competition intensity: when perceived idea qualities are low, the firm should set more dedicated resources for these projects or add extra shared resources to avoid excessive competition. When perceived idea qualities are medium, the firm should foster a competitive environment by reducing shared resources or increasing their scope and frequency of use. When perceived idea qualities are high, the firm should welcome any resource competition.**

Index Terms—**Adverse selection, incentive mechanism, moral hazard, parallel projects, product development.**

I. Introduction

IN PRACTICE, many firms follow the stage-gate process to manage product development from idea to launch [1]. A stage-gate process is composed of several stages divided by gates. The stages typically include new idea discovery, product design, and development. At the gate, the firm evaluates the project's progress and decides on project continuations and invested resources, known as the go/kill decision. Generally, the firm delegates stage execution to professional project managers. This delegation leads to the first agency issue (i.e., "moral hazard" issue) due to the unobservable nature of the manager's efforts. Therefore, in addition to investment decisions, the firm should design compensation contracts for project managers to incentivize their efforts.

Moreover, the firm lacks product development experience and thus may not be completely informed of each project manager's idea quality even after conducting project assessments [2]. By contrast, project managers usually possess superior idea quality information, and they may prioritize their own career, reputation, and experience gained from leading a project, which may conflict with the firm's interests. Therefore, they have incentives to overstate their idea quality, leading to the second agency issue known as "adverse selection." To dampen the managers' overstating incentives, the firm may reduce the investment in high-quality ideas when making investment decisions at the gate. It is evident that access to finance in the high-tech industry can be sharply curtailed due to information asymmetry, and the R&D investment is well below the level that would prevail if there were no market imperfections [3]. Moreover, many companies push up the bar for "go" decisions by setting a high hurdle rate of return for R&D projects [4], [5]. Furthermore, an inefficient investment scheme may discourage project managers from exerting necessary development efforts, complicating the effective compensation design. In short, the asymmetric information on idea quality and managers' efforts brings challenges to the firm in determining the effective investment scheme and compensation contract.

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The above challenges are particularly salient with parallel projects, where resource conflicts frequently arise and detrimentally impact project outcomes [6], [7], [8]. Parallel projects often compete against each other for the same resources, e.g., R&D budget, testing equipment, or specialists with unique expertise. This competition can lead to project delays or bottlenecks, exemplified in testing departments where project managers may queue for access to specialized laboratories [8], and in IT departments where skilled personnel are shared across multiple projects [7], [9]. From the firm's perspective, these resource conflicts complicate investment/resource allocation decisions, including which project to continue or which to obtain access priority, and the firm may encounter potential profit losses without proper investment schemes.

Consider the case of an innovation lab in Amsterdam that had invested in multiple innovation teams that tested various business ideas. Faced with budget cuts, the lab's leader, Duncan, encountered a dilemma that parallel projects were competing for limited R&D funding, and it was impossible to continue investing in all of them.[1] Moreover, the leader lacked information on the progress of the innovation teams toward finding business models that work. Therefore, selecting the right innovation projects was a daunting task for him.[2] Similar issues in portfolio decision-making arose in the European Food company. As noted by Kester et al. [10], in their portfolio meetings, a marketing manager would present a new product idea with supporting documentation. The executives often relied on managers' reports without fully understanding the proposed projects, i.e., they rarely scrutinized the data or questioned the project's underlying assumptions, even when the figures appeared unrealistic. This increased the risk of flawed investment decisions.

However, in the extant literature on stage-gate incentive design [2], [11], few of them consider parallel projects with resource conflicts (i.e., resource competition) and incorporate the design of the investment scheme. Our study aims to fill this gap by investigating the design of investment scheme and compensation contract in the presence of resource competition. The central research questions include:

1) How does the firm design an appropriate compensation contract for multiple managers to induce truthful idea quality information and encourage proper development effort at the same time?
2) What is the optimal investment scheme, and is there any investment inefficiency in the presence of adverse selection and moral hazard?
3) When should the firm allow resource competition among parallel projects to alleviate investment inefficiency?

To answer the above research questions, we build a principal-agent model where a risk-neutral firm employs two risk-neutral managers to lead their own projects. The firm follows a stage-gate process, which is composed of the research and development stages divided by a gate. At the research stage, the project managers propose their ideas and learn about the idea's quality (e.g., the technical feasibility), while the firm executives cannot obtain this idea quality information, resulting in adverse selection. Moral hazard exists in the late development stage, when the project managers determine whether to exert effort for product development. Between the two stages, the firm first designs the investment scheme and compensation contract that solicit project managers to truthfully report the idea quality and incentivize them to exert proper development efforts. Then, at the gate, project managers report their idea quality, and the investment decision for each project is specified based on the reported idea quality. The resources assigned to these projects may conflict, i.e., there may exist resource competition between the projects. After the development stage, the outcome of the continued project is realized, and the managers are compensated based on their project outcomes. We assume each project's outcome to be random, depending on the following factors: the idea quality, the development effort exerted by the manager, the firm's resource investment, and the resource competition.

Our results provide closed-form solutions for the investment scheme and compensation contract under the symmetric equilibrium. Under the optimal mechanism, the firm bases its investment decisions on the reported idea quality information. Specifically, the firm terminates low-quality ideas even though these ideas can bring positive returns with proper investments and continues medium- and high-quality ideas with low and high investments, respectively. As the idea quality increases, the firm should transition from fixed-salary contracts to outcome-based ones by offering higher unit bonuses but reduced fixed salaries. The rationale is as follows. The firm has incentives to invest more resources in higher-quality ideas due to the high marginal return on investment and offer a higher unit bonus as the high investment on a project increases the marginal return of the manager's effort. This provides incentives for the managers with lower quality ideas to overstate their idea quality. To counteract this motivation, the firm must pay information rent to those with higher quality ideas. To reduce the information rent, the firm tends to mitigate the overstating incentive by lowering investments and unit bonuses for higher quality ideas, referred to as the *overstatement-dampening* effect. However, the reduced investments and unit bonuses lead to lower efforts of the managers (called the *effort-discouraging* effect). The tradeoff between these two effects drives the optimal mechanism above.

Our result further shows that the investment in the presence of adverse selection and moral hazard is always below the first-best investment, referred to as the *underinvestment* issue. This issue is primarily driven by the asymmetric idea quality information and is exacerbated by unobservable efforts due to the tradeoff between the overstatement-dampening and effort-discouraging effects. Consequently, the firm should prioritize obtaining reliable data on project quality for efficient investment concerns. This underinvestment finding is in contrast to the overinvestment finding in the literature. In our setting, the project managers can adjust the effort levels based on the ideally known idea quality and have no incentive to put high effort into low-quality ideas, and thus, there exists no overinvestment issue.

Our model also underscores the impact of resource competition on the underinvestment issue. By comparing the degree of underinvestment with and without resource competition, we

[1] [Online]. Available: https://www.strategyzer.com/library/with-companies-cutting-budgets-what-should-innovators-do. The company's name cannot be revealed due to nondisclosure agreements (NDAs).

[2] [Online]. Available: https://www.strategyzer.com/library/to-kill-or-persevere-how-do-you-score-innovation-projects.

find that incorporating competition can help firms mitigate the underinvestment issue, and the firm should tailor the resource competition according to the perceived quality of the ideas. Specifically, the firm can improve investment efficiency by

1) allowing mild resource competition if the idea quality is likely to be low;
2) increasing the competition intensity if the idea quality is likely to be medium;
3) encouraging any resource competition if the idea quality is likely to be high.

The core rationale is that with high competition intensity, the project managers expect that a significant investment can severely impact the project return and, thus, have diminished incentives to overstate idea quality, which reduces the information rent and motivates the firm to invest more. This alleviation effect of competition on the underinvestment issue also plays a critical role in the firm profit. We find that the firm should either avoid project competition or allow a high degree of competition for a higher profit. This result is partly consistent with the intuition that firms should sharply limit the number of active projects and focus on central projects if the resource capacity is insufficient. Moreover, our result points out another possibility: high competition intensity may also benefit the firm profit in the presence of information asymmetry.

Finally, we show that our findings on the optimal mechanism are robust to the cases of risk-averse managers, limited resources, and more than two parallel projects. These cases undermine the investments compared to the base model.

To summarize, our contributions are threefold. First, in account of the stage-gate process with asymmetric idea quality information in the research stage and unobservable development effort in the late development stage, we characterize the optimal investment scheme and compensation contract for the parallel projects under resource competition. Second, we reveal that underinvestment inefficiency always exists in this process, primarily driven by asymmetric idea quality information, with unobservable effort exacerbating the issue. Third, we identify the adverse selection alleviation effect of resource competition and show that more intense resource competition may help alleviate this investment inefficiency and improve the firm's profit. Our findings offer guidelines for designing compensation contracts and investment schemes, and managing the intensity of resource competition among parallel projects to improve investment efficiency.

The rest of this article is organized as follows. We review the related literature in Section II. Section III presents the model. We investigate the optimal investment scheme and compensation contract in Section IV. Section V discusses the underinvestment issue. We present the robustness checks in Section VI. Finally, Section VII concludes this article. We relegate the proofs to Appendices, which are provided in the Supplementary Material.

II. Literature Review

This article studies the incentive design and investment scheme in product development with competing projects. Therefore, our work is related to the literature on incentive design in product development that focuses on R&D efficiency. Depending on the number of projects, the existing literature can be divided into single-project and multiproject settings.

The incentive design with the single-project setting has been extensively explored in the literature. Some of them focus on the effort incentive design under moral hazard, where the principal cannot observe the project manager's efforts that are critical to the project outcome [12], [13], [14], [15]. The project's inherent characteristics like project quality/performance and the agent's capability also affect the project outcome. As such information is usually the agent's private information and uncertain to the principal, there are studies examining the incentive design to motivate effort and induce truth-telling [16], [17], [18]. In addition to the incentive design, the firm needs to decide the resources allocated to the projects. However, the above research ignores the investment decision and the impact of asymmetric project information on the investment levels. Two related papers in the financial field, Harris and Raviv [19] and Bernardo et al. [20], addressed the capital investment problem of a principal that is uncertain of the technology productivity and project performance, respectively. They find that to induce truthful information, the firm encounters either overinvestment or underinvestment issues.

Similarly, the investment decision of NPD projects is also characterized by project uncertainty and decentralized decision-making associated with the stage-gate process (please refer to the review of the stage-gate process by Cocchi et al. [21]). Firms are uncertain about a project's quality/performance, which is the project manager's private information, when deciding whether to continue or abandon a project. In light of this uncertainty, firms, on the one hand, may continue underperforming projects, which can result in over featuring—products with more features than users actually need [22]. Both Levitt and Snyder [23] and Inderst and Klein [11] considered that the managers first create new investment opportunities by exerting effort at the research stage and then assist the firm in making the investment decision at the gate. They show that the firm may overinvest in unprofitable projects due to the high cost of idea discovery and the benefit of inducing high effort, respectively. On the other hand, firms may prematurely abandon projects with significant potential. Chao et al. [2] considered that the project manager learns about the project quality in the early research stage and then exerts effort in the subsequent development stage. The firm makes intervening go/kill decisions at the gate in between. They show that the firm underinvests by terminating some low-quality projects with positive value. Several studies have explored how to alleviate investment inefficiency. Hutchison-Krupat and Kavadias [24] compared the traditional stage-gate process with the bottom-up process in which the executive delegates the investment decision to the project manager, and find that the bottom-up process is efficient for more challenging projects. Oraiopoulos and Kavadias [25] and Hutchison-Krupat and Kavadias [26] examined the impacts of diverse decision-makers and advanced technology on investment efficiency. In addition to the decentralized feature, Bianchi et al. [27] showed that the linear feature of stage-gate is inadequate to deal with highly dynamic environments and leads to poor speed and cost performance. They recommend incorporating flexible agile models into the stage-gate process to improve the R&D performance.

In this article, we consider a similar stage-gate process as in Chao et al. [2]. In contrast, our model incorporates parallel projects and considers resource competition among them. We examine the investment scheme (including the "go/ kill" decisions) and the compensation contract by considering the interplay of asymmetric information and resource competition. Our result reveals that project competition can impact investment efficiency and the firm's profit.

In the multiproject setting, the existing literature studies how to manage a portfolio of projects, which can be competing or collaborative, through resource allocation. Chao and Kavadias [28] addressed the dynamic resource allocation problem over multiple candidate projects in a centralized decision-making setting. Under the decentralized decision-making setting with symmetric information between the firm and the project managers, Kanodia [29] and Gao et al. [30] investigated the firm's resource allocation strategy among collaborative projects and projects with market competition, respectively. Considering asymmetric information between the firm and the project managers, the literature examines the resource allocation strategy alongside incentive design to address the information asymmetry issue. Harris et al. [31] considered a firm produces outputs using intermediate products produced by multiple independent divisions. The intermediate product in each division is determined by the manager's private effort, the division's productivity, and the resources allocated by the firm. They design transfer-pricing-based incentive mechanisms to induce true productivity information and high efforts from the managers. In innovation contests with competing projects, the literature examines how the firm designs effort incentive design and resource allocation strategy based on projects' relative performance [32], [33], [34].

The following studies further consider efficiency issues accompanied by information asymmetry. Friebel and Raith [35] considered that the firm makes the investment decisions and designs incentives for two competing projects with quality uncertainty, where the investment decision is made after the project outcome is realized. They show that quality information asymmetry does not distort the resource allocation decision. However, if the resource allocation decision is made before the realization of the project outcome, the strategy can be distorted downward/upward. Ederer and Manso [36] empirically demonstrated an underinvestment result driven by the free-riding effect of other managers' explorative actions. Chao et al. [37] investigated the firm's funding authority and incentives for a single manager who is responsible for dynamically allocating resources to two projects. They show that the use of variable funding drives higher investment efficiency, and thus, higher portfolio performance.

A related paper, Schlapp et al. [38], showed overinvestment inefficiency in the stage-gate process with resource competition. Their setting differs from ours in two ways. First, they consider the process with a product evaluation stage followed by a gate. That is, their timing sequence is that managers make the effort decision before observing private signals about the project information, with the gate decision occurring after the evaluation stage. Second, they assume that higher effort enhances the precision of project information. As a result, the firm tends to overinvest in compensation to ensure sufficient managerial efforts for information acquisition. In contrast, we consider a different stage-gate structure with a "research" stage, followed by a gate, and then a "development" stage. That is, effort decisions in our framework are made after the project managers observe idea quality, with the gate decision intervening between the two stages. We find that this structure prevents overinvestment but introduces underinvestment inefficiency. This occurs because project managers who have private information about idea quality can strategically adjust their effort based on the observed quality. As a result, they lack incentives to exert high effort for low-quality ideas, preventing the firm from overinvesting.

To summarize, our study falls into the multiproject setting with asymmetric information. We consider a typical stage-gate process, a relevant context but overlooked in this literature stream, where the asymmetric idea quality information exists before the gate and the unobservable effort occurs afterward. Furthermore, we characterize the interaction of the parallel projects through resource conflicts and do not consider the project's relative performance. We show the underinvestment inefficiency driven by asymmetric information within this process. We identify the adverse selection alleviation effect of resource competition, demonstrating that intense competition can potentially alleviate this inefficiency and improve the firm profit.

III. MODEL

Consider a risk-neutral firm that employs two project managers indexed by $i = 1, 2$ to develop their own products. The two products are not necessarily related, and their developments follow a simplified stage-gate process, including the "research" stage (i.e., the discovery of ideas) and the subsequent "development" stage (i.e., the transformation of an idea into a product), with one gate in between. Between the two stages, the firm first designs the investment scheme and compensation contract to motivate the managers to exert proper effort and truthfully report their idea qualities. Then, at the gate, the firm decides whether and how much to invest in each project (i.e., go or kill) based on the reported idea quality.

A. Stage-Gate Process

In the "research" stage, project managers propose their ideas and become better informed about the idea's quality. The idea quality θ_i follows the distribution $F_i(\cdot)$, which is commonly known by the firm and the managers. Let f_i be the density function with support $[0, \bar{\theta}]$ where $\bar{\theta} < +\infty$ and $g_i(\theta_i) = f_i(\theta_i)/(1 - F_i(\theta_i))$ be the associated hazard rate function. Without loss of generality, we assume that $g_i(\theta_i)$ is increasing in θ_i. This monotonicity condition is commonly imposed in the agency literature (e.g., [2], [39], [40]). Manager i privately learns its own idea quality θ_i with certainty, while the firm cannot be completely informed of the idea quality due to the lack of product development experience. Therefore, the idea quality exists as a piece of asymmetric information between the manager and the firm at the end of the research stage. We refer to this idea quality uncertainty the firm faces as *idea risk* following Chao et al. [2]. After that, the firm decides and announces the investment scheme and the compensation contract.

At the gate, each manager reports his idea quality $\tilde{\theta}_i$ (which may differ from the true quality θ_i) to the firm upon the announced investment scheme and compensation contract. Based on the reported idea quality $\tilde{\theta}_i$ and the specified investment scheme, the firm makes the go/kill decision and, if go, allocates R_i (e.g., capital, collaboration team) to project i. We incorporate the kill decision by allowing $R_i = 0$, which means that project i is terminated.

During the subsequent "development" stage, if manager i receives a nonzero resource R_i, it will covertly exert effort $\tilde{e}_i \geq 0$ with a cost $c(\tilde{e}_i)$ to develop the proposed idea. We assume a quadratic effort cost $c(e) = \frac{\beta e^2}{2}$, where $\beta > 1$ is the effort-aversion parameter and is the same for both managers.[3]

The project outcome is realized after the manager's effort is invested. In addition to the idea quality and the manager's effort, the project outcome also depends on the firm's investment [41] and resource conflicts (i.e., resource competition) between the two projects, as mentioned in the introduction. We define a constant γ as the degree of resource conflicts between these two projects. A higher γ indicates more overlapped resources and more intense resource competition, and a zero γ implies the absence of resource conflicts, i.e., the two projects require totally different resources. The project outcome Y_i is given by

$$Y_i = R_i(\theta_i + \tilde{e}_i) + \gamma R_i(m_i - R_j) + \epsilon, \tag{1}$$

where the randomness $\epsilon \sim N(0, \sigma^2)$ denotes the *development risk*. The first term describes the impact of the inherent idea quality and manager's effort, while the second term characterizes the effect of resource competition. For the first term, the additive form $\theta + e$ is common in the literature. This term indicates the complementarity between the invested resource and the idea quality as well as the exerted effort. That is, a bad idea without effort cannot create value regardless of the resource invested, and any idea without investment brings nothing. On the other hand, resource conflicts generate both a positive effect denoted by $\gamma R_i m_i$ and a negative one denoted by $-\gamma R_i R_j$ on the project outcome. Here, m_i measures the potential effectiveness of the resources for project i. The positive effect term implies that the presence of competition brings efficiency in the utilization of invested resources when project managers transform their ideas into products because competition can enhance innovation efficiency [42], [43]. For example, competition for limited equipment drives teams to meticulously plan experiments and coordinate schedules, ensuring optimal utilization of resources and ultimately benefiting project outcomes. On the other hand, the negative effect of competition reflects that the outcome of project i can be adversely affected by the investment in the opposite project j due to resource conflicts.

[3] We require $\beta > 1$ to ensure the concavity of the firm's objective function and rule out infinite investment in case of $\gamma = 0$.

We summarize the above timeline of the stage-gate process in Fig. 1. Notably, project managers learn their private idea quality information, and then the firm designs the investment scheme and the compensation contract. Afterward, each manager reports his idea quality to the firm and exerts the development effort. The key feature of the information asymmetry here is that the adverse selection occurs before the go/kill decision, and the moral hazard emerges afterward.

B. *Incentive Mechanism*

In the above decentralized stage-gate process, each project manager has private information on the idea quality and exerts unobservable effort in developing their ideas. To encourage the managers to truthfully report the idea quality and exert proper effort, the firm has two instruments in the incentive design: a compensation contract and an investment scheme.

Following the standard incentive in the new product development, we assume a linear compensation contract, i.e., $w(\tilde{\theta}, Y) = a(\tilde{\theta}) + b(\tilde{\theta})Y$. Both the fixed salary $a(\tilde{\theta})$ and the unit bonus $b(\tilde{\theta})$ in the outcome-based component hinge on the reported idea quality. For the second instrument, the invested resource $R(\tilde{\theta})$ is also based on the reported idea quality.

The firm incurs the investment cost $\frac{R^2}{2}$, which is quadratic in the investment volume [44], [45]. In innovation-driven industries, R&D costs often escalate nonlinearly due to the high marginal costs required to secure additional resources. A quadratic cost function captures this dynamic, reflecting how the pursuit of innovation leads to disproportionately higher expenses. For example, firms may bid up wages for highly skilled specialists and spend heavily on cutting-edge infrastructure to maintain progress in R&D breakthroughs. The firm's payoff, denoted by Π, is, thus, the expected project outcome minus the wages paid to the managers and the investment cost

$$\Pi(w, R) = \sum_{i=1,2} \mathbb{E}\left[Y_i - w(\tilde{\theta}_i, Y_i) - \frac{R(\tilde{\theta}_i)^2}{2}\right].$$

For manager i, given the true idea quality θ_i, the payoff from reporting $\tilde{\theta}_i$, and exerting effort $\tilde{e}_i$ is the expectation of the utility function $u(\cdot)$ over the monetary wage plus the conceptual project value subtracting the effort cost

$$U_i(\tilde{\theta}_i, \tilde{e}_i | \theta_i) = \mathbb{E}\left[u\left(w(\tilde{\theta}_i, Y_i) + \delta_i \theta_i R(\tilde{\theta}_i) - c(\tilde{e}_i)\right) \Big| \theta_i\right].$$

Regarding the conceptual project value, the coefficient $\delta_i \in [0, 1]$ measures manager is preference for the idea quality and the investment originating from its reputation concern [19]. For example, the manager gains accomplishment and accesses better future opportunities due to the experience of leading an excellent project with a great idea and costly investment. We restrict $\delta_i \leq 1$ to reflect the fact that the firm values the project and investment more than the project managers. Besides, we assume that manager i faces an outside option $\underline{U}_i \geq 0$.

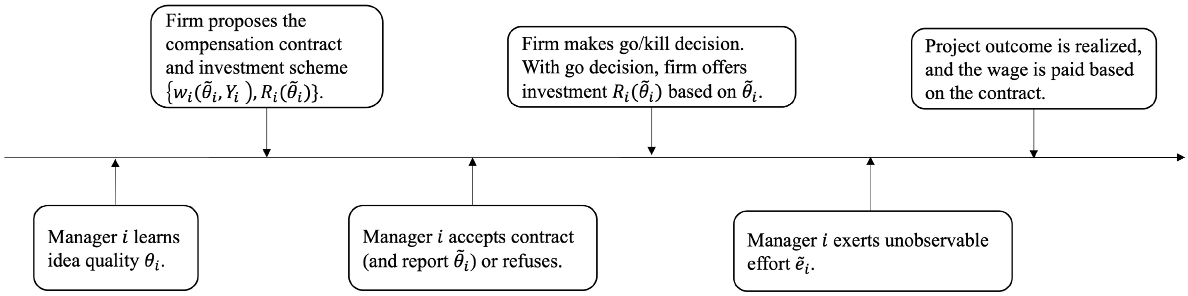


Fig. 1. Sequence of events.

IV. Optimal Incentive Mechanism Design

Confronted with asymmetric idea quality information, the firm designs the compensation contract $w(\tilde{\theta}, Y)$ and the investment scheme $R(\tilde{\theta})$ to motivate the managers to report the true idea quality. By the revelation principle [46], it is without loss of generality to restrict our attention to direct mechanisms in which the manager reports the idea quality truthfully.

A. Incentive Design Problem

To start with, we first investigate the optimal effort level of manager i who leads a project with idea quality θ_i but reports $\tilde{\theta}_i$. In the base model, we assume both managers are risk-neutral. For analytical traceability, we consider the symmetric Nash equilibrium for project managers.[4] Specifically, manager i makes decisions given the other manager's best response, i.e., truthfully reporting the idea quality θ_j and exerting the optimal effort e_j.

Recall that, manager is payoff when exerting effort $\tilde{e}_i$ is given by

$$\begin{aligned} &U_i(\tilde{e}_i|\theta_i, \tilde{\theta}_i, \tilde{\theta}_j) \\ &= \mathbb{E}\left[w(\tilde{\theta}_i, Y_i) + \delta_i\theta_i R(\tilde{\theta}_i) - c(\tilde{e}_i)\middle|\theta_i\right] \\ &= \mathbb{E}\left\{a(\tilde{\theta}_i) + b(\tilde{\theta}_i)\left[R(\tilde{\theta}_i)(\theta_i + \tilde{e}_i) + \gamma R(\tilde{\theta}_i)\right.\right. \\ &\quad \left.\left.\left(m_i - \gamma R(\tilde{\theta}_j)\right) + \epsilon\right] + \delta_i\theta_i R(\tilde{\theta}_i) - \frac{\beta\tilde{e}_i^2}{2}\right\}. \end{aligned}$$

Since $U_i(\tilde{e}_i|\theta_i, \tilde{\theta}_i, \tilde{\theta}_j)$ is strictly concave in $\tilde{e}_i$, the optimal effort level, denoted by e_i, can be obtained via the first-order condition as follows:

$$e_i(\tilde{\theta}_i) = \frac{b(\tilde{\theta}_i)R(\tilde{\theta}_i)}{\beta}. \tag{2}$$

Note that the optimal effort level does not depend on the true idea quality. Instead, it depends on the reported $\tilde{\theta}_i$ via the unit bonus $b(\tilde{\theta}_i)$ and the invested resource $R(\tilde{\theta}_i)$. It indicates that the high investment and unit bonus encourages the manager to exert more effort.

[4] Such an equilibrium has been widely considered in the contest literature [33], [38], [47], [48]. In the first three studies, manager i makes the effort decision given other managers' best effort decisions. In Schlapp et al. [38], manager i makes decisions given that the other one exerts high effort and truthfully reports its private information.

Given that the other manager $j = \{1, 2\}\backslash i$ truthfully reports the true idea quality θ_j, substituting this optimal effort $\tilde{e}_i = e_i(\tilde{\theta}_i)$ into the payoff function $U_i(\tilde{e}_i|\theta_i, \tilde{\theta}_i, \theta_j)$ gives the maximal payoff of manager i who reports the idea quality $\tilde{\theta}_i$, denoted by $U_i(\tilde{\theta}_i|\theta_i, \theta_j)$. Hence, $U_i(\tilde{\theta}_i|\theta_i, \theta_j) \equiv \arg\max_{\tilde{e}_i} U_i(\tilde{e}_i|\theta_i, \tilde{\theta}_i, \theta_j) = U_i(e_i(\tilde{\theta}_i)|\theta_i, \tilde{\theta}_i, \theta_j)$. Then,

$$\begin{aligned} U_i(\tilde{\theta}_i|\theta_i, \theta_j) &= a(\tilde{\theta}_i) + b(\tilde{\theta}_i)R(\tilde{\theta}_i) \\ &\left[\theta_i + \frac{b(\tilde{\theta}_i)R(\tilde{\theta}_i)}{\beta} + \gamma(m_i - R(\theta_j))\right] \\ &+ \delta_i\theta_i R(\tilde{\theta}_i) - \frac{(b(\tilde{\theta}_i)R(\tilde{\theta}_i))^2}{2\beta}. \end{aligned}$$

Given the managers' optimal effort levels, the firm expects to design an incentive mechanism that maximizes its expected payoff. As mentioned above, we focus on the direct mechanisms in which the managers report the idea quality truthfully. Then, the firm's problem can be stated as follows:

$$\max_{a(\theta_i), b(\theta_i), R(\theta_i)} \sum_{i=1,2} \mathbb{E}\left[Y_i - a(\theta_i) - b(\theta_i)Y_i - \frac{R(\theta_i)^2}{2}\right] \tag{3}$$

$$\begin{aligned} \text{subject to} \quad & \theta_i \in \arg\max_{\tilde{\theta}_i \in [0, \bar{\theta}]} U_i(\tilde{\theta}_i|\theta_i, \theta_j) \quad \forall\theta_i \in (0, \bar{\theta}), \\ & i = 1, 2, \qquad \sim\sim\sim[(\mathrm{IC}_i^Q)] \\ & U_i(\theta_i|\theta_j) \geq \underline{U}_i,\ i = 1, 2, \qquad [(\mathrm{IR}_i)] \\ & R(\theta_i) \geq 0,\ b(\theta_i) \geq 0. \end{aligned}$$

The incentive-compatible condition IC_i^Q ensures that manager i truthfully reports the idea quality given manager j reports the true idea quality, i.e.,

$$U_i(\theta_i|\theta_j) \geq U_i(\tilde{\theta}_i|\theta_i, \theta_j) \quad \forall\theta_i, \tilde{\theta}_i, \theta_j \in (0, \bar{\theta}),\ i = 1, 2$$

,where $U_i(\theta_i|\theta_j) \equiv U_i(\theta_i|\theta_i, \theta_j)$ is manager is maximal payoff when truthfully reporting the idea quality and exerting the optimal effort. The manager's individual-rationality condition IR_i ensures its participation in product development.

Recall that we consider the symmetric equilibrium, the parameters for the two managers should be identical, and thus, we drop the subscript i in the following analysis, i.e., $\delta_i = \delta_j = \delta$, $m_i = m_j = m$, $\underline{U}_i = \underline{U}_j = \underline{U}$, $f_i(\cdot) = f_j(\cdot) = f(\cdot)$.

B. Optimal Compensation Contract and Investment Scheme

To isolate the role of unobservable effort (moral hazard) and asymmetric idea quality information (adverse selection), we first provide the optimal mechanism under cases of symmetric information and pure moral hazard below, followed by the case of both adverse selection and moral hazard.

Lemma 1:

1) In the symmetric information case where the idea quality is known to the firm and the manager's effort is observable, the first-best investment scheme and compensation contract are

$$R^B = \frac{\beta[(1+\delta)\theta + \gamma m]}{\beta - 1 + 2\gamma\beta},\ a^B = \underline{U},\ b^B = 0.$$

Moreover, the first-best effort level is $e^B = R^B/\beta$.

2) When the idea quality is known to the firm, but the managers' effort is unobservable (i.e, there is only moral hazard), the optimal mechanism is

$$R^{MH} = \frac{\beta[(1+\delta)\theta + \gamma m]}{\beta - 1 + 2\gamma\beta},$$
$$a^{MH} = \underline{U} - \left[\theta + \left(\frac{1}{2\beta} - \gamma\right) R^{MH} + \gamma m + \delta\theta\right],$$
$$b^{MH} = 1.$$

Besides, the optimal effort level is $e^{MH} = R^{MH} b^{MH}/\beta$.

Lemma 1 indicates that with symmetric idea quality information, it is always optimal to continue projects with positive investment. The investment decision increases in the idea quality due to the boosting effect of high idea quality on the marginal return of investment. Moreover, in the first-best case, the firm only pays a fixed salary that is equal to the outside option to meet the project manager's participation constraint. When the firm cannot observe the effort level, it pays a positive unit bonus to incentivize effort from the project manager.

When the idea quality information becomes unknown to the firm (i.e., with moral hazard and adverse selection), the firm solves program (3) for the optimal mechanism. Formally, let θ^1 and θ^2 be the solution to $g(\theta)[(1+\delta)\theta + \gamma m] - \delta = 0$ and $g(\theta)[(1+\delta)\theta + \gamma m] - \delta - (1+2\gamma)\beta = 0$, respectively; it follows that $\theta^1 < \theta^2$.[5] Denote $\check{\theta} \equiv \max\{0, \theta^1\}$ and $\hat{\theta} \equiv \max\{0, \theta^2\}$. The optimal mechanism is described as follows.

Proposition 1 (Optimal mechanism): With unobservable effort and asymmetric idea quality information, the optimal investment scheme, denoted by $R^\star$, and the compensation contract, denoted by $a^\star$ and $b^\star$, depend on the idea quality as follows.

1) *Project-termination region* $\theta \in [0, \check{\theta}]$*:* The firm terminates low-quality projects and compensates the managers for the outside option, i.e., $R^\star = 0, a^\star = \underline{U}, b^\star = 0$.

2) *Fixed-salary region* $\theta \in [\check{\theta}, \hat{\theta}]$*:* The firm continues medium-quality projects with investment

$$R^\star = \frac{g(\theta)[(1+\delta)\theta + \gamma m] - \delta}{g(\theta)(1+2\gamma)}$$

and pays each manager with $a^\star = \underline{U} - (\delta\theta R^\star - \int_{\check{\theta}}^{\theta} \delta R^\star(s) ds) < \underline{U}$, $b^\star = 0$.

3) *Linear-contract region* $\theta \in [\hat{\theta}, \bar{\theta}]$*:* The firm continues high-quality projects with investment

$$R^\star = \frac{\beta\left[g(\theta)\left((1+\delta)\theta + \gamma m\right) - (1+\delta)\right]}{g(\theta)(\beta - 1 + 2\gamma\beta)}$$

and pays each manager with

$$a^\star = \underline{U} - \left[(b^\star + \delta)R^\star\theta - \int_{\check{\theta}}^{\theta} (b^\star(s) + \delta) R^\star(s) ds\right] - b^\star R^\star \left(\frac{b^\star R^\star}{2\beta} + \gamma m - \gamma R^\star\right) < \underline{U},$$

$$b^\star = 1 - \frac{\beta - 1 + 2\gamma\beta}{g(\theta)[(1+\delta)\theta + \gamma m] - (1+\delta)} \in (0, 1).$$

Proposition 1 describes the optimal investment scheme and compensation contract across the idea quality. Compared with the symmetric idea quality information cases, the optimal investment under adverse selection depends on the distribution of idea quality. Specifically, the firm refrains from investing in low-quality ideas, even if they could yield positive returns with proper investment as in the symmetric information case. In contrast, the firm continues investing in medium- and high-quality ideas. Moreover, as the idea quality improves, the compensation scheme shifts from compensating the managers for their outside option to a fixed salary smaller than the outside option and eventually to an outcome-based contract. This result explains the prevalence of performance-based contracts in practice and underscores the importance of adjusting fixed salaries based on idea quality. Furthermore, it indicates that firms should prioritize resource allocation toward higher quality ideas, aligning with empirical evidence from field studies in high-tech companies [49].

Under this optimal mechanism, the profit from low-quality ideas is negative due to the compensation for the managers' outside option. As a result, the optimal expected profit is lower than that under symmetric information as the investment deviates from the first-best efficiency.

To better understand the optimal mechanism, we present the comparative statics analysis regarding the idea quality as follows.

Proposition 2 (Impact of idea quality): For the compensation contract, the unit bonus $b^\star$ is nondecreasing in the idea quality θ while the fixed salary $a^\star$ is nonincreasing in θ. Besides, the investment $R^\star$ is nondecreasing in the idea quality θ.

Built upon Proposition 1, Proposition 2 states that under the optimal mechanism, the firm allocates more investment $R^\star$ and offers a higher unit bonus $b^\star$ but a lower fixed salary $a^\star$ to the manager who reports a higher idea quality. This result contrasts with the moral hazard case where $b = 1$ is independent of the idea quality. Notice that the firm has incentives to invest more resources in higher quality ideas due to the higher marginal return of investment and offer a higher unit bonus to the managers as the high investment on a project increases the marginal return of a manager's effort. As a result, given idea quality θ, any manager

[5] This relationship follows from the increasing failure rate property of $g(\theta)$, which implies that $\theta g(\theta)$ is increasing in θ. In addition, we assume that $g(\bar{\theta})[(1+\delta)\bar{\theta} + \gamma m] - \delta - (1+2\gamma)\beta \geq 0$ to ensure $\theta^2 \leq \bar{\theta}$.

of lower quality idea $\theta' < \theta$ has the incentive to overstate as θ to secure the higher investment and unit bonus for the higher project outcome and reputation concern.[6]

To counteract such motivation, the firm must pay information rent to the idea with quality θ. To reduce the information rent, the firm tends to mitigate the overstating incentive by lowering the investment and unit bonus for the idea with quality θ compared to the moral hazard case. By doing so, a manager with θ' finds it worse to overstate its quality as the increase in the outcome-based salary and the reputational gain cannot compensate for the loss in fixed salary and the higher effort cost. We refer to the positive effect of reduced investment and unit bonuses in discouraging the upward imitation as the *overstatement-dampening* effect. However, the reduced investment and unit bonus inevitably lead to a lower effort level, as indicated by the optimal effort, which negatively impacts the project outcome and harms the firm. This negative consequence is termed the *effort-discouraging* effect.

The property of the optimal mechanism stems from the tradeoff between the overstatement-dampening effect related to the asymmetric idea quality information and the effort-discouraging effect associated with the unobservable development effort. Specifically, for projects with high-quality ideas, the effort-discouraging effect dominates due to the high marginal return of investment and the managers' efforts. Therefore, the firm should invest substantial resources and offer generous unit bonuses to boost the effort and project outcome, even if bearing the information rent. For projects with medium-quality ideas, the marginal return of investment and the managers' efforts decrease. In this case, the overstatement-dampening effect becomes significant, and the firm offers zero unit bonus to reduce the information rent but still offers positive investment for the positive project return concern. For projects with low-quality ideas, their marginal return of investment and the managers' efforts are small, and thus, the overstatement-dampening effect is remarkably significant. To reduce the information rent, the firm further decreases the investment and unit bonus for these projects to zero. Meanwhile, to ensure that each type of project manager obtains the truth-telling utility, the firm increases the fixed salary as the idea quality decreases.

Next, we explore the effect of the resource competition intensity γ on the optimal mechanism by analyzing how the three regions and the associated compensation contract as well as investment scheme change. Note that the optimal contract and investment in the project-termination region are constant given the outside option. Let θ^3 and θ^6 be the solution to $\theta g(\theta) - \delta/(1+\delta) = 0$ and $\theta g(\theta) - 1 = 0$, respectively.

Proposition 3 (Impact of resource competition intensity):

1) The project-termination region shrinks as the resource competition intensity γ increases if $\gamma \leq \frac{\delta}{g(0)m}$, and disappears otherwise.
2) For the fixed-salary region, this region expands as the resource competition intensity γ increases if $m \leq \frac{2\beta}{g(\theta^2)}$ and disappears if $m > \frac{2\beta}{g(0)}$ and $\gamma > \frac{\delta+\beta}{g(0)m-2\beta}$. Besides, $R^\star$ decreases in γ if and only if $\theta > \theta^3$ and $m \leq \frac{2[\theta g(\theta)(1+\delta)-\delta]}{g(\theta)}$.
3) For the linear-contract region, this region shrinks as the resource competition intensity γ increases if and only if $m \leq \frac{2\beta}{g(\theta^2)}$. Besides, both $R^\star$ and $b^\star$ decrease in γ if and only if $\theta > \theta^6$ and $m \leq \frac{2\beta(1+\delta)(\theta g(\theta)-1)}{g(\theta)(\beta-1)}$.

As resource competition intensifies, the project-termination region narrows, and it eventually disappears when the competition gets extremely intense. This is because, in such conditions, substantial investment in competing projects is likely to harm the outcome of any given project, especially given the project with high quality. To mitigate this negative impact, the firm may tend to restrict the investment in those projects. Notably, reducing investments in high-quality ideas diminishes the incentives for the managers with lower quality ideas to overstate their project's quality. In other words, intense resource competition helps lower the information rent, an effect referred to as the *adverse selection alleviation* effect of resource competition. Due to this alleviation effect, the firm tends to switch from zero to positive investment in low-quality ideas; that is, the project-termination region shrinks.

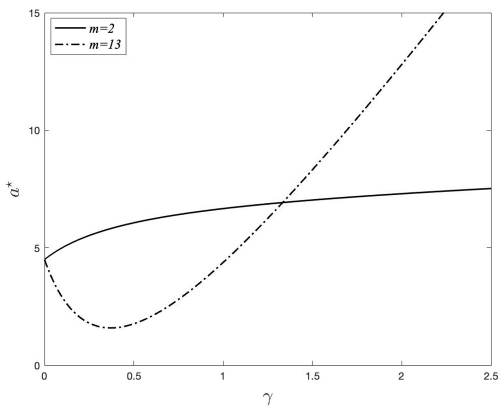
In the fixed-salary and linear-contract regions for higher quality ideas, the adverse selection alleviation effect strengthens. Intuitively, one might expect the firm to increase investment with more intense competition, but the optimal mechanism depends on the magnitude of m (see Table I for detailed region changes). A large m indicates a significant marginal positive effect of competition, and thus, a high investment can still improve project outcomes. As γ rises, the firm increases both the investment and unit bonus as they are complementary for effort. This expands the linear-contract region. In addition, the fixed-salary region widens only if the project-termination region narrows more sharply. When m is small, the increased investments may harm project outcomes due to the limited positive marginal effect. Then, the firm lowers investment and unit bonus as γ increases, narrowing the linear contract region. In this case, the fixed-salary region expands as the other two shrink. This result suggests that the firm should carefully offer unit bonuses with a linear contract under resource competition. When resources are frequently shared (i.e., high γ) and competition's benefit is limited (i.e., small m), providing a unit bonus may not be optimal.

We also numerically illustrate the monotonicity of $a^\star$ in γ in Fig. 2(a). In general, $a^\star$ behaves in the opposite direction to $R^\star$ and $b^\star$. With limited investment effectiveness m, $a^\star$ increases with γ. As m increases, $a^\star$ becomes nonmonotonic in γ. This is because the fixed salary must balance the information rent for the manager's truth-telling utility and the compensation that the managers obtain. When m is small, the firm reduces both the investment and unit bonus as γ increases. While this reduction lowers the information rent, it diminishes the manager's overall compensation more due to the reduced project outcome. To maintain the truth-telling incentive, the firm compensates by increasing the fixed salary. Conversely, the firm increases the investment and unit bonus as γ increases. This results in a tradeoff: while the manager's information rent increases, so does

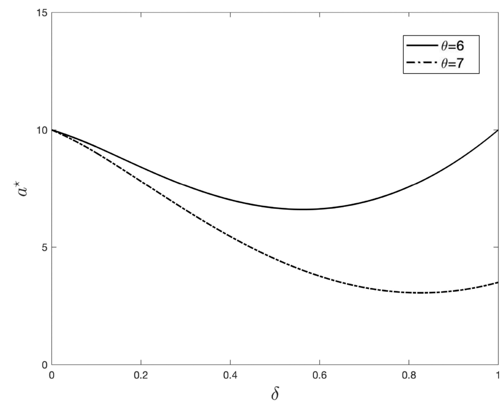
[6]Notice that the project outcome is increasing in the investment as long as the investment is not extremely substantial.

TABLE I
IMPACT OF RESOURCE COMPETITION INTENSITY ON THE THREE REGIONS (PT: PROJECT-TERMINATION REGION, FS: FIXED-SALARY REGION, LC: LINEAR-CONTRACT REGION)

Condition		Regions change when γ increases
$m \le \frac{2\beta}{g(\theta^2)}$	$\gamma \le \frac{\delta}{g(0)m}$	PT $\downarrow$, FS $\uparrow$, LC $\downarrow$
	$\gamma > \frac{\delta}{g(0)m}$	PT disappears, FS $\uparrow$, LC $\downarrow$
$m \in (\frac{2\beta}{g(\theta^2)}, \frac{2\beta}{g(0)}]$	$\gamma \le \frac{\delta}{g(0)m}$	PT $\downarrow$, FS is nuanced, LC $\uparrow$
	$\gamma > \frac{\delta}{g(0)m}$	PT disappears, FS $\downarrow$, LC $\uparrow$
$m > \frac{2\beta}{g(0)}$	$\gamma \le \frac{\delta}{g(0)m}$	PT $\downarrow$, FS is nuanced, LC $\uparrow$
	$\gamma \in (\frac{\delta}{g(0)m}, \frac{\delta+\beta}{g(0)m-2\beta}]$	PT disappears, FS $\downarrow$, LC $\uparrow$
	$\gamma > \frac{\delta+\beta}{g(0)m-2\beta}$	PT & FS disappear, LC $= (0, \bar{\theta})$



(a)



(b)

Fig. 2. Impact of γ and δ on the fixed salary ($\beta = 2, \theta \sim U$[0,20]). (a) $\theta = 6, \delta = 0.5$. (b) $m = 2, \gamma = 1$.

their total compensation, leading to the observed nonmonotonic pattern. With less intense competition, the manager's compensation grows faster than the information rent, prompting the firm to lower the fixed salary to reduce costs. In contrast, with more intense competition, the information rent rises more sharply, driving up the fixed salary.

Proposition 3-(1) and (2) further indicate that, when the managers show less preference for the invested resources (i.e., a small δ), the project-termination or fixed-salary regions are more likely to disappear as the competition intensifies. This is because a smaller δ reduces the manager's incentive to overstate, allowing the firm to invest without significantly limiting investment to reduce the information rent.

In the following result, we present more comparative statics for the optimal investment scheme and compensation contract concerning the effectiveness of the resources m, the effort-aversion parameter β, and the manager's preference for the invested resources δ.[7] Let $\hat{\beta}$ be the solution of (13) in Appendix I (given in the Supplementary Material).

[7]We omit straightforward results for instruments unaffected by parameters.

Corollary 1:

1) In the fixed-salary region, $R^\star$ increases in m, and increases in δ if and only if $\theta > \theta^6$. Moreover, $a^\star$ is nonincreasing in m.
2) In the linear-contract region, $R^\star$ and $b^\star$ increase in m and decrease in β. Similar to the fixed-salary region, both are increasing in δ if and only if $\theta > \theta^6$. Moreover, $a^\star$ is nonincreasing in m, and is increasing in β if and only if $\beta \le \max\{1, \hat{\beta}\}$.

An increase of m increases the marginal benefit of investment, and the firm would like to increase the investment. This further improves the marginal benefit of the effort, and thus, the firm also increases the unit bonus. The high m, coupled with the elevated investment and unit bonus, benefits the project outcome as well as the manager's compensation, thus, the firm can reduce the fixed salary to save the compensation cost. A small β decreases the marginal cost of effort, making the firm more desirable to induce high effort from the managers. To this end, the firm would increase the investment and the unit bonus. As β decreases, the manager's compensation first increases at a slower rate than the information rent as the manager has to pay the effort cost. To ensure the truth-telling utility, the firm should increase the fixed salary. When β decreases a lot, the manager's compensation increases at a sharper rate as the manager's effort cost declines a lot, which makes the fixed salary decrease.

The manager's preference for the invested resources (i.e., δ) has two effects. On the one hand, a high δ indicates that the managers have high preferences for the invested resources and are more likely to overstate the idea quality, resulting in a high information rent. On the other hand, a high δ recoups the manager's utility so that the firm can reduce the compensation to ensure the truth-telling utility. For the ideas with relatively low quality, the former effect dominates, and the firm would reduce the investment and unit bonus to reduce the information rent. For the ideas with relatively high quality, the latter effect dominates, and the firm would increase the investment and unit bonus to ensure a higher outcome. As for the fixed salary, we find that $a^\star$ can be nonmonotonic in δ, as shown in Fig. 2(b). This property can be again attributed to the balance between the increased information rent and compensation for the manager.

V. Underinvestment Issue

As shown in the optimal mechanism, the firm invests nothing in low-quality ideas and encounters a negative profit. This indicates that the optimal investment scheme may be insufficient for the projects to bring positive returns, and the firm should seriously account for such inefficiency. In this section, we analyze the investment inefficiency issue. We say that *underinvestment* exists when the investment $R^\star$ is below the first-best investment R^B. The underinvestment issue is said to be severe if the investment gap, denoted by $\Delta R \equiv R^B - R^\star$, is large. In what follows, we first analyze this investment efficiency issue from the perspective of idea quality information asymmetry and then examine the effect of idea quality and resource competition.

A. Role of Idea Risk

Recall that whether the managers' effort is observable or not does not affect the investment decision, as shown in the benchmark cases. In contrast, the idea quality uncertainty alters the firm's investment decision, as stated in Theorem 1.

Theorem 1 (The role of idea risk): Underinvestment occurs when the idea quality is unknown to the firm, i.e., $\Delta R(\theta) > 0, \forall\theta$.

Theorem 1 shows the primary role of adverse selection in generating underinvestment, as the firm must distort its investment decision to reduce the information rent associated with the asymmetric idea quality information. It indicates that the firm fails to provide sufficient investment to develop a project, which explains the commonly observed underinvestment issue in high-tech R&D [3], and the high hurdle rate in product development process [4], [5]. Furthermore, it indicates that the firm should prioritize obtaining reliable data on project quality to alleviate underinvestment. As indicated in the innovation lab case, the consulting group recommends using scorecards to assess the evidence supporting a project's desirability, viability, and adaptability.

Interestingly, we can show that the investment under pure adverse selection is between the first-best level and that under moral hazard and adverse selection [the proof under pure adverse selection is provided in Appendix II (given in the Supplementary Material)]. From this angle, we say that the unit bonus serves as a primary tool to incentivize efforts from the managers, and the investment can be leveraged to induce truth-telling of the asymmetric idea quality information. With both adverse selection and moral hazard, the firm is required to balance the overstatement-dampening and effort-discouraging effects and adjust the unit bonus and investment simultaneously. Specifically, although the firm is supposed to use $b = 1$ and $R = R^B$ to induce the highest effort, it has to reduce both due to the overstatement-dampening effect. The smaller unit bonus b indicates a lower effort level, and consequently, a lower marginal return of investment, encouraging the firm to reduce R further. Hence, the investment level deviates even further from the first-best level. To summarize, the interaction of adverse selection and moral hazard exhibits a mutual intensification effect, exacerbating the distortion of the investment scheme.

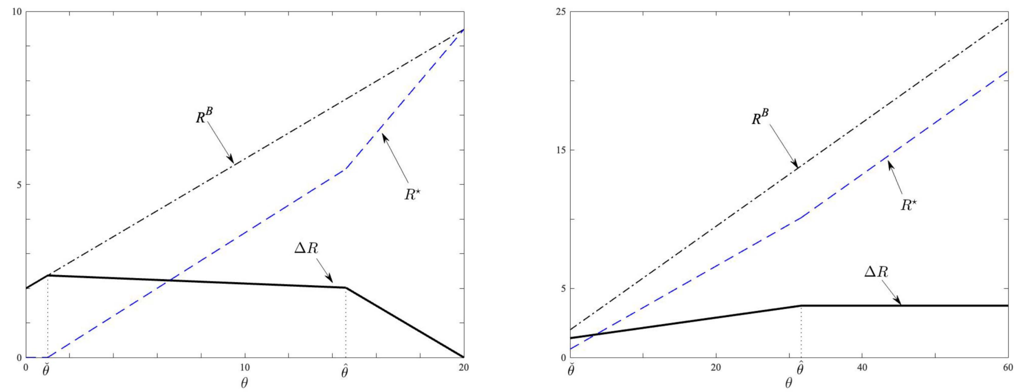
Our underinvestment findings contrast with the overinvestment inefficiency in Schlapp et al. [38], which investigates a stage-gate process featuring a product evaluation stage preceding a gate. In their framework, project managers decide their effort before receiving private signals about project information; moreover, because higher effort is assumed to improve the precision of these signals, firms tend to overcompensate to secure sufficient managerial effort. In contrast, our model adopts a different stage-gate structure, where effort decisions are made after project managers observe idea quality and after the gate decision. This sequence enables project managers, who hold private information on idea quality, to adjust their effort accordingly, thus eliminating the incentive to exert high effort on low-quality ideas and preventing the firm from overinvestment. This disparity underscores the role of process timing in shaping investment schemes. During the early periods (e.g., following idea proposal), the firm should allocate less investment than the first-best level at the gate. Conversely, at the gate in later periods (e.g., after market or technology evaluation), a higher investment than the first-best level should be guaranteed.

The following result further shows how the underinvestment issue depends on the idea quality.

Proposition 4: The severity of underinvestment is not monotonic in the idea quality θ. Specifically:

1) for low idea quality ($\theta \in [0, \check{\theta}]$), the investment gap ΔR is increasing in θ;
2) for high idea quality ($\theta \in [\hat{\theta}, \bar{\theta}]$), the investment gap ΔR is decreasing in θ;
3) for medium idea quality ($\theta \in [\check{\theta}, \hat{\theta}]$), the monotonicity of the investment gap depends on the distribution of θ. Specifically, ΔR is increasing in θ when $\frac{g'(\theta)}{(g(\theta))^2} < \frac{1+\delta}{\delta(\beta-1+2\gamma\beta)}$ while decreasing when $\frac{g'(\theta)}{(g(\theta))^2} > \frac{1+\delta}{\delta(\beta-1+2\gamma\beta)}$.

Intuitively, when the idea quality is high, the high marginal return of investment indicates that the firm is reluctant to deviate from the first-best investment. As such, the underinvestment issue is less severe as the idea quality improves for high-quality



(a) (b)

Fig. 3. Impact of the idea quality distribution on the investment decisions ($m = 4, \delta = 0.5, \beta = 1.1, \gamma = 2, E[\theta] = 10$). (a) $\theta \sim U$[0,20]. (b) $\theta \sim$ Exp($\frac{1}{10}$).

ideas. However, the investment gap is not always decreasing when the idea quality exhibits low and medium levels (see the solid line in Fig. 3). Recall that, under the optimal mechanism, adverse selection twists the firm's investment decision for low-quality ideas to zero. In this case, the investment gap equals the first-best investment level and, thus, is increasing in the idea quality. For medium-quality ideas, the tradeoff between the profit loss due to underinvestment and the overstatement-dampening effect is more nuanced. Thus, the monotonicity of the gap depends on the distribution of θ that affects the information rent. As illustrated in Fig. 3, by fixing all the other parameters, ΔR is decreasing in θ for $\theta \in [\check{\theta}, \hat{\theta}]$ if θ follows the uniform distribution while ΔR increases with θ if θ follows the exponential distribution.

Project uncertainty is widely acknowledged as a critical constraint on the R&D performance. For instance, Bianchi et al. [27] demonstrated that the linearity feature of stage-gate models can impede innovation velocity and escalate costs in highly dynamic environments. They advocate for the integration of flexible agile approaches—such as consumer involvement and frequent testing—into the stage-gate process. While their work highlights the drawbacks of inflexible features, our study focuses on the organizational aspects of stage-gate systems, particularly the interaction between decentralized decision-making and project uncertainty. We identify how this structural feature can lead to underinvestment. In the following, we discuss potential strategies firms might adopt to mitigate this inefficiency.

B. Role of Resource Competition

Recall that resource competition imposes two opposite effects on the project outcome. We are interested in whether the competition can alleviate or amplify this underinvestment issue. To this end, we compare the investment gap ΔR with competition (i.e., $\gamma > 0$) to the counterpart without competition (i.e., $\gamma = 0$). Note that $\gamma = 0$ implies that there are two independent projects, which replicates the typical setting of one firm and one manager in the incentive design literature [2], [20]. Denote θ^4 as the solution to $(1+\delta)\theta g(\theta) - (\delta + \beta) = 0$. It can be verified that $\check{\theta} < \theta^3 < \theta^4$ by definition.

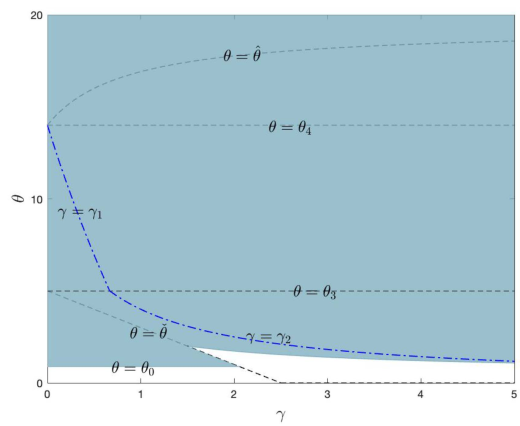


Fig. 4. Impacts of θ and γ on underinvestment ($m = 4, \delta = 0.5, \beta = 3, \theta \sim U[0, 20]$).

Let $\theta^0 = \frac{m(\beta-1)}{2\beta(1+\delta)}$, $\gamma^0 = \min\left\{\frac{\delta}{mg(\theta^0)} - \frac{\beta-1}{2\beta}, \frac{\delta}{g(0)m}\right\}$, $\gamma^1 = \frac{\beta-1}{2\beta}\left(\frac{\beta(1+\delta)}{\theta g(\theta)(1+\delta)+\delta(\beta-1)} - 1\right)$, and $\gamma^2 = \frac{\beta-1}{2\beta}\left(\frac{1}{\theta g(\theta)} - 1\right)$. In the following result, we provide sufficient conditions under which resource competition can alleviate the underinvestment issue.

Theorem 2 (The role of resource competition in underinvestment): The existence of resource competition can alleviate the underinvestment issue if one of the following conditions is satisfied:

1) if the idea quality is low and the competition intensity γ is small ($\theta^0 < \theta \leq \check{\theta}, \gamma \leq \gamma^0$);
2) if the idea quality is medium and the competition intensity γ is large ($\check{\theta} < \theta \leq \theta^3$, $\gamma \geq \gamma^2$ or $\theta^3 < \theta \leq \theta^4$, $\gamma \geq \gamma^1$);
3) if the idea quality is high ($\theta \in (\theta^4, \bar{\theta}]$).

Theorem 2 shows that the allowance of resource competition among the parallel projects may improve the investment efficiency, depending on the idea quality and competition intensity. We further depict the impacts of resource competition and idea quality on the magnitude of underinvestment in Fig. 4, where the shaded area indicates that underinvestment is alleviated. Clearly,

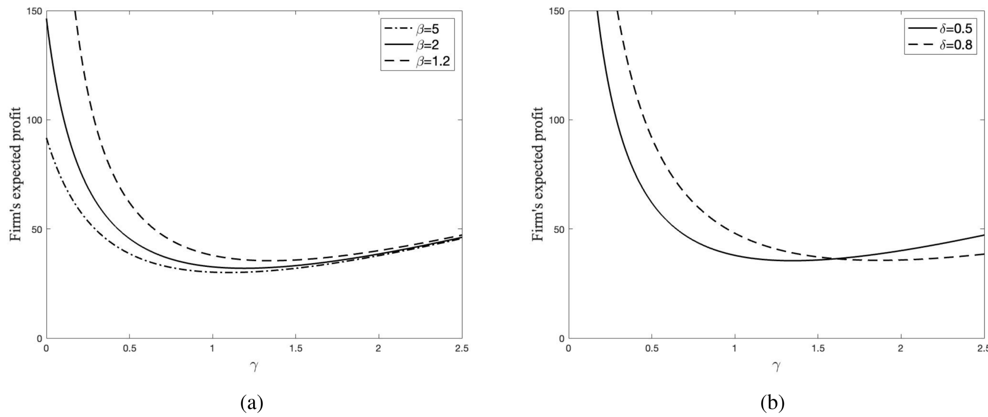


Fig. 5. Impact of γ on the firm's expected profit. (a) $\delta = 0.5$. (b) $\beta = 1.2$.

the shaded areas confirm the sufficiency of the conditions in Theorem 2.

For low-quality ideas with $\theta^0 < \theta \leq \check{\theta}$, the investment is distorted to zero under adverse selection. In this case, the first-best investment with mild resource competition ($\gamma < \gamma^0$) is smaller than that without competition, indicating that mild resource competition can mitigate the underinvestment issue. For medium-quality ideas, the firm always provides positive investment. The investment efficiency can be improved when the resource competition is intense enough such that the adverse selection alleviation effect is significant to inhibit the project managers' overstatement incentives. For high-quality ideas $\theta \in [\theta^4, \bar{\theta}]$, the adverse selection alleviation effect becomes more prominent. In addition, as the overstatement-dampening effect and the resulting underinvestment issue are mild (see Proposition 4), even a low competition intensity can mitigate the underinvestment issue, thereby improving investment efficiency. This result indicates that firms can leverage resource competition to improve investment efficiency, especially for those concerned with high hurdle rates in R&D. As noted in Paul et al. [50] and Xie et al. [43], increasing the number of undergoing projects and effectively managing highly intensified tasks can improve R&D efficiency. Our result provides more operational insights on how to tailor the resource competition to improve investment efficiency. Specifically,

1) if the idea qualities of the parallel projects are more likely to be low, the firm should set more dedicated resources for these projects or add extra shared resources to avoid excessive competition. More specifically, the firm can provide additional guidance for these projects, arrange more functional managers to share knowledge, or implement strict funding criteria to ensure these projects receive the necessary resources;
2) if the idea qualities of the parallel projects are more likely to be medium, the firm should foster a competitive environment in R&D by reducing shared resources or increasing the scope and frequency of resources that need to be shared. For example, the firm can develop a shared equipment pool where teams must apply for priority access, rotate access to experts among parallel projects, and allow competition among projects with similar targets/technology;
3) if the idea qualities of the parallel projects are more likely to be high, firms should actively encourage resource competition.

C. Impact of Resource Competition on the Firm Profit

In this part, we numerically illustrate the impact of resource competition γ on the firm's expected profit in Fig. 5. The parameters are set as: $\theta \sim U[0, 20]$, $m = 4$, $\underline{U} = 10$. We are also interested in how this impact depends on the manager's effort cost coefficient β and the reputational value of the project δ.

As we can see, the firm's profit is first decreasing and then increasing in γ. One immediate implication of this observation is that the firm should avoid moderate resource competition between projects. Part of this observation is consistent with the intuition that firms should sharply limit the project number and focus on the central project if the resource capacity is insufficient. However, our result also points out another possibility: highly intense competition may improve the profit. This can be again attributed to the adverse selection alleviation effect of resource competition. As shown in Theorem 2, when the resource competition intensifies, it improves the investment efficiency for projects with relatively high qualities.

VI. Robustness Check

In this section, we check the robustness of the optimal mechanism by considering three cases: risk-averse managers, limited resources, and N parallel projects.

A. Risk-Averse Managers

In this section, we consider the risk-averse managers [2], [48], [51]. Note that, under the risk-neutral assumption, the development risk denoted by the variance σ^2 does not affect the compensations and investments. This risk-averse scenario allows us to further investigate how the development risk affects these decisions.

For analytical simplicity, we focus on the manager is certainty equivalent. Note that maximizing the expected utility of manager i is equivalent to maximizing their certainty equivalent. Given the true idea quality θ_i and the reported $\tilde{\theta}_j$ from the opponent manager j, manager is certainty equivalent from exerting the optimal effort e_i and reporting $\tilde{\theta}_i$ is

$$\begin{aligned}\mathrm{CE}(\tilde{\theta}_i \mid \theta_i, \tilde{\theta}_j) &= \max_{e_i}\{u^{-1}(U_i)\} \\ &= \max_{e_i}\{u^{-1}(\mathbb{E}[u(w_i(\tilde{\theta}_i, Y_i) + \delta_i\theta_i R_i(\tilde{\theta}_i) - c(e_i))|\theta_i, \tilde{\theta}_j\end{aligned}$$

where $u(x) = -e^{-\eta x}$, and $\eta > 0$ represents the degree of risk aversion. This negative exponential utility function is widely used in the principal-agent literature [2], [52]. We obtain the symmetric equilibrium as follows.

Proposition 5: Under the risk-averse manager case, the optimal compensation contract and investment scheme under the symmetric equilibrium are given by the following.

1) *Project-termination region ($\theta \in [0, \breve{\theta}]$):* The firm terminates the projects (i.e., $R^A = 0$), and pays the managers with $a^A = \underline{U}$, $b^A = 0$.
2) *Fixed-salary region ($\theta \in [\breve{\theta}, \hat{\theta}]$):* The firm continues the project with investment $R^A = \frac{g(\theta)[(1+\delta)\theta+\gamma m]-\delta}{g(\theta)(1+2\gamma)}$, and pays the manager with $a^A = \underline{U} - (\delta\theta R^A - \int_{\breve{\theta}}^{\theta} \delta R^A(s)ds)$, $b^A = 0$.
3) *Linear-contract region ($\theta \in [\hat{\theta}, \bar{\theta}]$):* The firm continues the project with investment $R^A = R_1$, where R_1 solves $[(1+\delta)\theta + \gamma m]g(\theta) - (1 - \frac{R_1(\beta R_1 + 2\eta\sigma^2 g(\theta)\beta + R_1^2 g(\theta))}{(\eta\sigma^2\beta + R_1^2)\beta})\frac{(R_1 g(\theta)-\beta)R_1}{(\beta\eta\sigma^2+R_1^2)g(\theta)} - \delta - (1 + 2\gamma)g(\theta)R_1 = 0$. The firm pays each manager with $a^A = \underline{U} - [(b^A + \delta)R^A\theta - \int_{\hat{\theta}}^{\theta}(b^A(s) + \delta)R^A(s)ds] - b^A R^A[\frac{b^A R^A}{2\beta} + \gamma(m - R^A)] + \frac{(b^A)^2\eta\sigma^2}{2}$, $b^A = \frac{(R^A g(\theta)-\beta)R^A}{[\beta\eta\sigma^2+(R^A)^2]g(\theta)}$.

Proposition 5 confirms our main results in Proposition 1 that the firm would terminate low-quality ideas and only continue medium- and high-quality ideas. It is intriguing to find that the bounds of the three regions are independent of the development risk σ^2, remaining the same as the risk-neutral case. Moreover, the optimal mechanisms in the project-termination and fixed-salary regions are independent of σ^2. This is because the development risk affects the firm's payoffs through the term b, and the effect of σ^2 disappears due to zero b.

In the linear-contract region, we show the impact of development risk σ^2 on the investment R^A through the numerical experiment (see Fig. 6). We assume $\theta \sim U[0, 20]$, $m = 4$, $\beta = 2$, $\eta = 1$, $\gamma = 1$, $\delta = 0.5$. Note that the development risk σ^2 emerges in the development stage and reduces the payoffs of both the managers and the firm. For any fixed idea quality θ, a higher development risk indicates a lower unit bonus, impeding the effort of the project managers. The reduced effort, in turn, decreases the investment.

B. Limited Resources

As noted in the innovation lab example, the firm may face budget cuts for R&D. We assume that parallel projects are

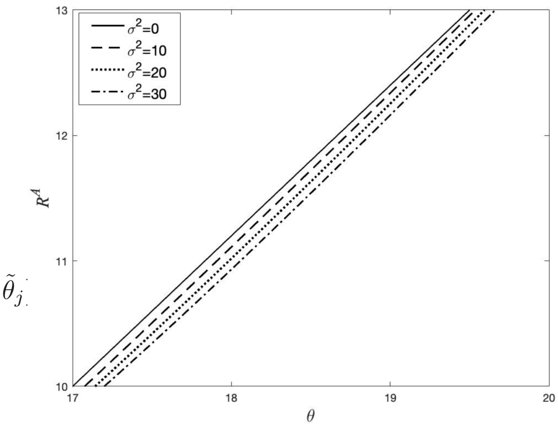


Fig. 6. Impact of θ and σ^2 on R^A in the linear-contract region.

subject to the budget constraint $\bar{R}$. The firm's optimization problem is defined in program (3) with an additional constraint $\sum_{i=1,2} R_i \le \bar{R}$. The symmetric equilibrium is presented in Table III of Appendix I (given in the Supplementary Material). To differ from the base model, we impose the condition $\bar{R} \le \frac{2\beta[(1+\delta)\bar{\theta}+\gamma m]}{\beta-1+2\gamma\beta}$. Otherwise, the budget constraint does not affect the optimal mechanism.

Compared to the base model without the budget constraint, we find that the optimal mechanism structure in Proposition 1 preserves. The investment for relatively low-quality ideas is within the budget and remains the same as in the base model. In contrast, for relatively high-quality ideas, the investment is bounded from above by $\frac{\bar{R}}{2}$ and, thus, lower than that without the budget constraint. A smaller budget increases the range of idea quality constrained by available resources. Therefore, the investment is further limited as the ceiling gets lower.

C. N Parallel Projects

To aid clarity, we have presented a model with two projects. However, the underlying tradeoff between the overstatement-dampening and effort-discouraging effects is a general phenomenon that survives when we allow more than two parallel projects. Consequently, the optimal mechanism should be preserved. In particular, the project outcome becomes $Y_i = R_i(\theta_i + \tilde{e}_i) + \gamma R_i(m_i - \sum_{j\neq i} R_j) + \epsilon$ when there are N parallel projects. We can see that more parallel projects intensify the marginal negative effect of resource competition on the project outcome. In light of this effect, the firm may tend to lower the investment compared to the base model.

VII. Conclusion

For effective management of parallel projects, firms often follow the stage-gate process, which includes the research stage for idea discovery, the late development stage for product development, and a gate in between. This decentralized process accompanies the adverse selection issue on asymmetric idea quality information and the moral hazard issue on unobservable development efforts for the firm. In addition, resource conflicts among these parallel projects may complicate the design of investment schemes and compensation contracts. In a setting that incorporates the above challenges, we find that the tradeoff between the overstatement-dampening and effort-discouraging effects determines the optimal investment scheme and compensation contract. Specifically, the firm should terminate low-quality ideas even if they can generate positive returns upon proper investment due to the significant overstatement-dampening effect. Conversely, as the idea quality increases, firms should continue the projects and increase the investments accordingly. In addition, it is optimal to provide more unit bonuses to the managers who report higher quality ideas and reduce their fixed salary.

We show that the asymmetric idea quality information can create an underinvestment situation in which the potential projects fail to secure sufficient investment, and the unobservable development effort can further exacerbate this situation. This result explains why many companies maintain high hurdle rates in R&D. This finding contrasts with the overinvestment finding in previous studies with alternative stage-gate processes, underscoring the role of process timing in shaping investment schemes. Specifically, the firm should commit less investment than the first-best level at early-period gates (e.g., following idea proposal stage), while higher investment—exceeding the first-best level—is warranted at later-period gates (e.g., after market or technology evaluation). If underinvestment in early periods significantly undermines profitability, firms can leverage scorecards to obtain reliable project quality data for alleviating underinvestment.

We further highlight the operational details of how idea risk and resource competition affect investment efficiency. We identify that the adverse selection alleviation effect of resource competition may help improve investment efficiency, and the firm should adjust competition intensity based on perceived idea quality. Specifically, if the idea qualities of the parallel projects are more likely to be low, the firm should set more dedicated resources for these projects or add additional shared resources to avoid excessive competition. If the idea qualities of the parallel projects are more likely to be medium, the firm should foster a competitive environment in R&D by reducing shared resources or increasing the scope and frequency of resources that need to be shared. If the idea qualities of the parallel projects are more likely to be high, the firm should welcome any resource competition.

For future research avenues, it might be interesting to examine settings where the effort cost for each manager is affected by the competition intensity, e.g., high competition intensity leads to a higher development cost for the manager to achieve the same project outcome.

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